MULTIFAN-CL to opal: south-west Pacific albacore

Bridging the 2024 WCPFC south Pacific albacore diagnostic model, a two-region MULTIFAN-CL model, to a single-region opal model

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1 Overview

This document converts the 2024 stock assessment of south Pacific albacore (*Thunnus alalunga*) [1], a two-region MULTIFAN-CL [2], [3] model, into opal, an RTMB age-structured model with one region. Any model runs here are preliminary explorations and should not be used *as is* for management advice. opal is not changed: where it cannot represent the MFCL model, the MFCL model is simplified (in MFCL) or the difference is measured with replays (standalone simulations) and documented. The comparisons are meant as advice to opal's developers on the features the stock would need.

- **An MFCL bridge to model region 1** (Section 3). Four steps, each a refitted MFCL model: the diagnostic model (M0), no movement (M1), region 1 only (M2), and growth fixed without the conditional age-at-length data (M3). Region 1′s spawning biomass roughly doubles at each step: the stock's scale is weakly determined, and each source of information removed lets it drift up.
- **The conversion** (Section 4) maps MFCL's October-September model year, its quarterly and monthly fishing incidents, the indices and the length compositions onto opal with four seasons, and takes biology at age from MFCL.
- **At MFCL's values opal reproduces MFCL** (Section 5): its initial numbers and recruitment exactly, its spawning biomass within 2.1%. Replays put the gap on catch timing (start-of-quarter removal against MFCL's Baranov removal at each incident's date); seasonal selectivity hardly matters for spawning biomass.
- **The fits** (Section 6). With MFCL's selectivity fixed, opal's scale is unbounded (O1): the length compositions keep improving towards an unexploited stock. The cause (Section 7) is length-at-age timing: MFCL predicts each length sample at its date with lengths that grow through the year, opal from start-of-year length-at-age. With opal's growth at MFCL's mid-year lengths (O1-mid) the fit converges at MFCL's scale.

- **Seasonal set-ups** (Section 8). Splitting the fisheries with seasonal selectivity into one fishery per season, and an opal model with quarterly age classes and time steps (recruitment kept annual), are set up through opal's inputs alone.
- **What opal would need** is summarised in Section 9.

2 The 2024 assessment

The assessment [1] covers the south Pacific from the equator to 50°S and from 140°E to the coast of South America, in two regions: region 1 is the southern WCPFC Convention Area, including the overlap area with the EPO, and region 2 the rest of the EPO. Fisheries are defined as areas-as-fleets within each region (six areas in region 1, three in region 2).

- **Time.** 1954 to the third quarter of 2022. The model year runs October-September with recruitment on 1 October (week 37 of the calendar year in the report), 12 annual age classes (the last a plus group); catches are taken by quarter (longline, driftnet) or month (troll).
- **Fisheries.** 17 extraction fisheries (13 in region 1: longline by flag group and area, two troll fisheries and a driftnet fishery; 4 in region 2) and three index fisheries (region 1: longline north of 25°S and New Zealand troll; region 2: longline). Catches are in numbers except troll and driftnet (weight). The catch-conditioned approach solves the Baranov catch equation for fishing mortality at each fishing incident.
- **Indices.** sdmTMB standardised CPUE [4] with time-varying CVs; the troll index's generic CV is set to 0.2.
- **Length compositions.** 50 2-cm bins from 30 cm, re-weighted by catch; robust normal likelihood; sample sizes capped at 1000 and divided by Francis (2011) weights; samples below 50 fish dropped.
- **Conditional age-at-length.** 654 otolith ages in 33 fishery-quarter strata, effective sample size 0.75 of the observed.
- **Biology.** von Bertalanffy growth with offsets at ages 2-4 (L_1 fixed at 45.5 cm); M-at-age inversely proportional to length [6], scaled to average 0.36 over ages 4-12; spawning potential from maturity and sex ratio at length; Beverton-Holt steepness 0.8 with a weak penalty (CV 0.71) on recruitment deviations; fixed recruitment distribution (0.819 in region 1) and movement from SEAPODYM [7].
- **Selectivity.** Length-based cubic splines (3-4 nodes), grouped across fisheries (16 groups), seasonal (four seasons) for the tropical longline fisheries 1-7 and the EPO fishery 14, non-decreasing for the longline indices.

The assessment found the stock not overfished and not subject to overfishing (median $SB_{recent}/SB_{F=0}$ 0.48 across the model ensemble, including estimation uncertainty).

3 The MFCL bridge

Table 1: The MFCL bridge (M0 diagnostic, M1 no movement, M2 region 1 only, M3 growth fixed without age-at-length). Objective (negative log-likelihood plus penalties) and its components; region 1 spawning (adult) biomass in thousand tonnes; depletion in 2022 against the unfished projection. M0′s region 1 values are from the two-region model.

step	objective	length	age-length	CPUE	max grad.	SB 1954	SB 2022	depletion
M0	−41,824.6	−42,457.3	786.4	−170.0	5.5e-07	871	251	0.502
M1	−41,813.2	−42,455.2	788.3	−162.3	3.3e-06	1,716	375	0.567
M2	−36,315.1	−37,009.1	792.2	−111.8	6.2e-07	3,413	460	0.617
M3	−37,149.1	−37,052.2		−112.7	7.4e-06	6,592	859	0.751

The diagnostic model was supplied with MFCL 2.2.7.0, which reproduces its objective function value exactly (M0). Each later step is refitted with the diagnostic model's phase 9 settings (2200 evaluations), warm-started from the previous step; a fit takes 15-20 minutes. M1 and M3 stop at the evaluation limit with the line search stalled, at maximum gradients of a few times 10^{-6}.

- **M1, no movement.** The (fixed, SEAPODYM-derived) movement matrices are set to the identity, so the regions are closed populations. They still share the recruitment deviations (total recruitment split 0.819 / 0.181) and the stock-recruitment relationship, so the EPO data still inform region 1′s recruitment.
- **M2, region 1 only.** The EPO fisheries (14-17) and EPO index (20) are dropped and the region 1 indices renumbered 14 and 15. The par file is built from M1′s: fishery- and region-indexed values are subset, region 1′s recruitment is kept by shifting the total population scaling parameter by log(0.818971), and the sections shaped by regions, movement or catch records come from MFCL's `-makepar` for the one-region files (all zero or structural in this model). M2 removes the recruitment link to the EPO.
- **M3, growth fixed, no age-at-length.** opal has no conditional age-at-length likelihood, so growth is fixed at M2′s estimates and the data dropped.

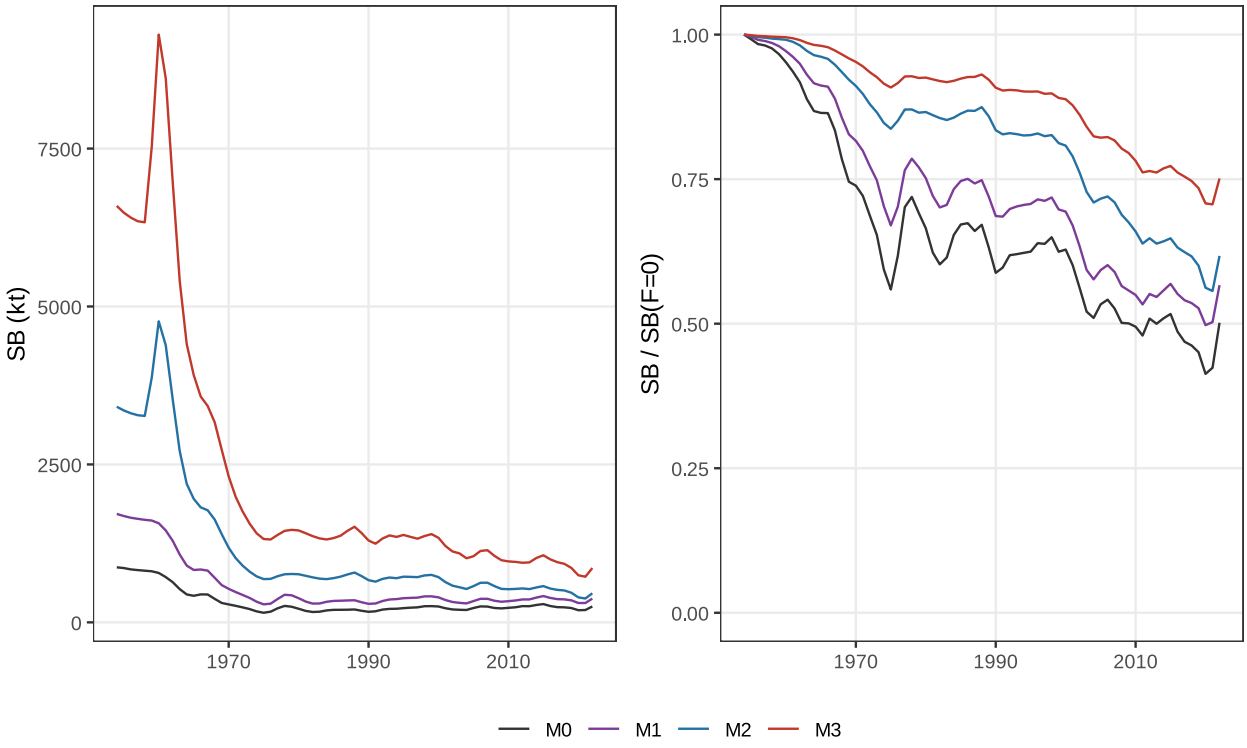


Figure 1: Region 1 spawning biomass and depletion along the MFCL bridge.

Region 1′s spawning biomass in 1954 goes from 871 kt (M0) to 1,716, 3,413 and 6,592 kt, and depletion in 2022 from 0.50 to 0.75. In this catch-conditioned model the scale comes from the indices, the length compositions and their interaction with the catch; movement, the shared recruitment with the EPO and the age-at-length data each constrain it, and it drifts up as each is removed.

Table 2: Transplant check: M2′s one-region structure evaluated at M1′s estimates (no refit) against M1′s region 1. Largest relative difference.

quantity	max rel. diff.
recruitment	0.0e+00
numbers at age (1954)	3.0e-03
numbers at age (all years)	3.7e-02
adult biomass	1.3e-02
predicted catch (extraction fisheries)	0.0e+00
selectivity at age	0.0e+00
mean length at age	0.0e+00
natural mortality at age	0.0e+00

With no movement, region 1 in M1 is unaffected by region 2′s catch or abundance (tested by setting region 2′s catch and recruitment near zero), so the transplanted model should reproduce it. Recruitment, selectivity, growth, M and the predicted catches agree exactly, but the numbers at the oldest ages drift up by a few percent from a small (0.02% per age) difference in the initial age structure, and adult biomass by up to 1.3%. The cause, somewhere in MFCL's one-region structure, is not identified; M2 is refitted, so it does not carry into the bridge.

4 Conversion to opal

`shared/mfcl-to-opal/` reads MFCL's `.par`, `.frq`, plot report and `length.fit` in base R and builds opal's inputs (`mfcl_to_opal()`).

Table 3: Fisheries in opal (M3). MFCL diagnostic: the fishery number in the two-region model. Seasons: distinct selectivity patterns by season in MFCL. Length data: the fish flag 49 divisor of the capped sample sizes, the number of quarters and of model years with compositions.

fishery	MFCL diagnostic	name	catch units	catch total	seasons	divisor	quarters	years
1	1	1 LL DWFN 1a	numbers	5,238,575	4	80	0	0
2	2	2 LL DWFN 1b	numbers	14,751,211	4	183	0	0
3	3	3 LL DWFN 1c	numbers	29,906,078	4	51	178	54
4	4	4 LL DWFN 1d	numbers	21,609,922	4	164	145	49
5	5	5 LL PICT 1ab	numbers	2,383,218	4	287	0	0
6	6	6 LL PICT 1ab	numbers	31,848,058	4	89	116	31
7	7	7 LL AU/NZ 1ab	numbers	535,380	4	29	25	13
8	8	8 LL DWFN 1cd	numbers	56,708,825	1	372	89	42
9	9	9 LL PICT 1cd	numbers	1,718,893	1	212	17	14
10	10	10 LL AU/NZ 1cd	numbers	4,421,636	1	125	74	27
11	11	11 TR 1c	weight (t)	108,687	1	27	74	25
12	12	12 TR 1d	weight (t)	48,506	1	37	14	7
13	13	13 DN 1cd	weight (t)	41,679	1	358	2	2
14	18	14 LL index 1abcd	numbers	0	1	746	57	57
15	19	15 TR index 1ef	numbers	0	1	55	25	25

- **Time.** MFCL's model year starts in calendar month 10: the `.frq` dimension line's ninth field (MFCL's `month_1`), which the report gives as week 37. Calendar months 10-12 belong to the next model year (so the data's last model year, 2022, has no calendar October-December). opal has 69 model years of four seasons (October-December, January-March, ...), with recruitment and ageing at the start of the year. Fishing incidents in the same fishery and season are summed (troll catches are monthly).
- **Catch** is in numbers for the longline fisheries and in tonnes for troll and driftnet; the two indices remove nothing.
- **Indices.** MFCL's survey likelihood is log-normal with variance $\lambda_i \sigma^2$, where λ_i is the `.frq` CV column over its mean and σ fish flag 92 / 100 (0.2): opal's standard errors are $\sigma\sqrt{\lambda_i}$. Both indices are annual (longline in July-September, troll in January-March).
- **Length compositions.** Samples below 50 fish are dropped, each is rescaled to its sample size capped at 1000, and they are summed by fishery and model year, with MFCL's divisors as opal's variance adjustments, under opal's multinomial likelihood (MFCL: robust normal per incident).
- **Biology.** Mean weight, maturity and M at age from MFCL's report, so opal's spawning biomass is MFCL's adult biomass (numbers x weight x maturity at the start of the year). opal's growth is used only for the length compositions: von Bertalanffy with age-specific K multipliers at ages 1-4 for MFCL's independent mean lengths at ages 2-4 (RMSE 0.52 cm), SD linear in mean length between MFCL's SDs at ages 1 and 12 (largest difference 0.47 cm).
- **Selectivity.** MFCL's selectivity-at-age, averaged over seasons weighted by each season's catch (index fisheries: their survey season), as opal's `sel_fa_external`.

5 opal at MFCL's values

O0 holds everything at MFCL M3's values. `mfcl_match_recruitment()` sets opal's B0 from MFCL's Beverton-Holt curve ($R_0 = \alpha - \beta/\mathrm{SPR}_0$, $B_0 = R_0\mathrm{SPR}_0$), its initial deviations so that opal starts from MFCL's 1954 numbers-at-age, and its recruitment deviations (bias adjustment off) so that it recruits MFCL's recruitment every year.

Table 4: O0 against MFCL M3: largest relative differences.

quantity	max rel. diff.
numbers at age, first year	4.8e-07
recruitment	2.0e-14

The replays are standalone simulations with MFCL's estimates (not opal code). They cross catch removal at the start of each quarter, then a quarter of M (opal), or at each fishing incident's date with M and F acting together until the next incident (MFCL's catch-conditioned Baranov), with MFCL's season-averaged selectivity (opal) or its seasonal selectivity.

Table 5: Replays: spawning biomass against MFCL M3 (relative difference in 2022 and largest annual) and against O0.

timing	selectivity	rel. 2022	max |rel.|	max |rel.| vs O0
opal	average	1.56%	2.10%	0.06%
opal	seasonal	1.60%	2.16%	0.19%
mfcl	average	−0.31%	0.55%	2.32%
mfcl	seasonal	−0.26%	0.53%	2.26%

The opal-timing replay with averaged selectivity reproduces O0, which checks the replay. Catch timing accounts for almost all of the gap between O0 and MFCL; seasonal selectivity changes spawning biomass by less than 0.1%. The MFCL replay is within 0.6% of MFCL; the remainder is largest in the mid-1990s, after the driftnet years, and may be the weight-at-age MFCL uses for catches in weight (not confirmed).

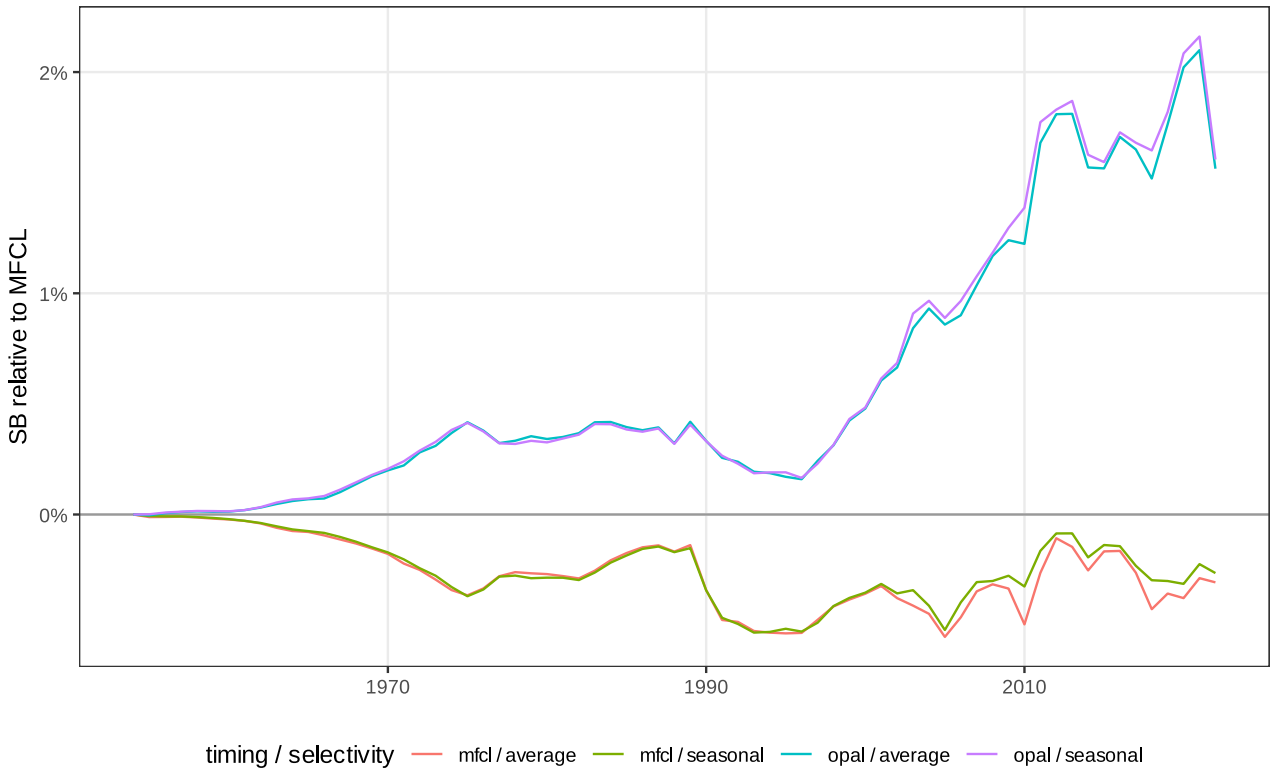


Figure 2: Spawning biomass relative to MFCL M3: O0 and the replays.

6 opal fits

O1 holds MFCL's season-averaged selectivity, growth, steepness (0.8), sigma_R (0.64, MFCL's CV of 0.71), M and the biology, and estimates B0, the recruitment deviations for 1955-2020 (the last two at the curve, as MFCL sets them at the mean), and the two index catchabilities. The initial population is unfished. O1 runs to the upper bound of log(B0) (22.00 against MFCL's 14.04): spawning biomass in the billions of tonnes and depletion near 1.

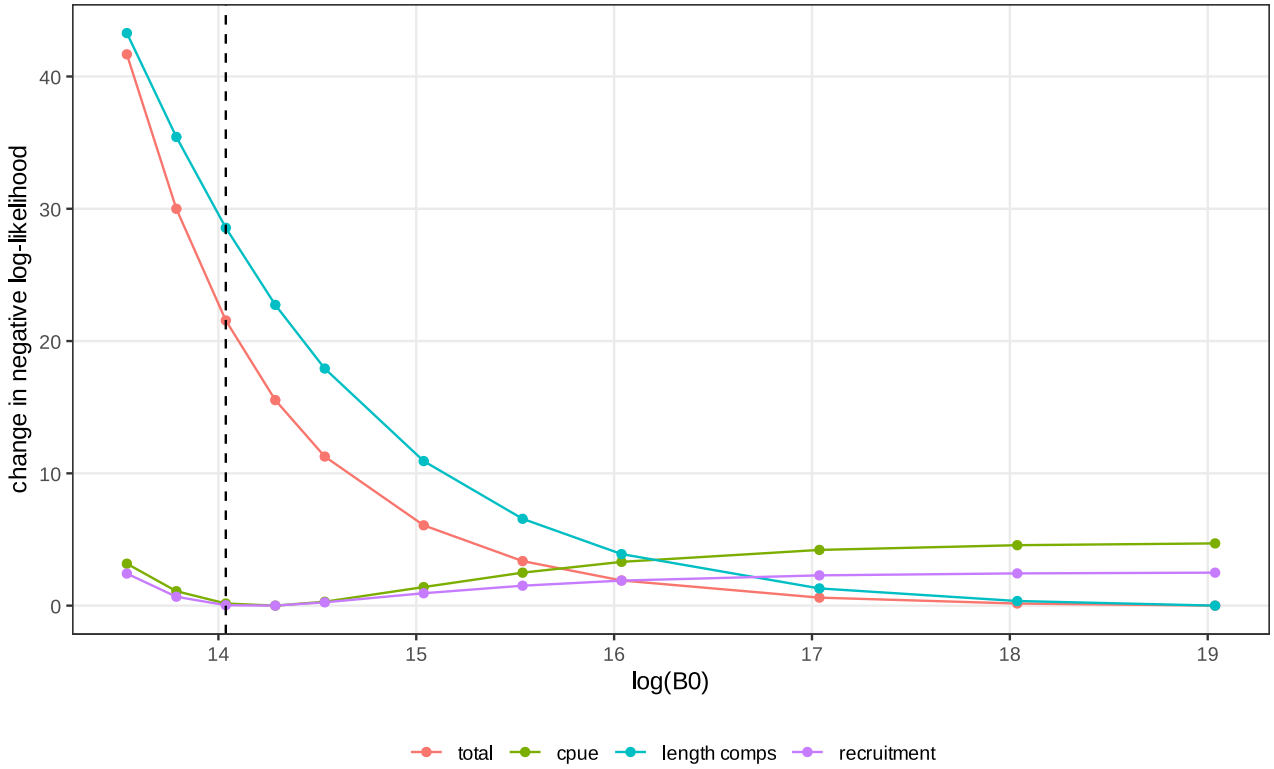


Figure 3: Profile over log(B0) for O1 (recruitment deviations and catchabilities estimated at each value): change in the objective and its components from their smallest values. The dashed line is MFCL's scale.

The index and the recruitment penalty are best near MFCL's scale; the length compositions keep improving towards an unexploited stock and win.

7 Why the length compositions favour an unexploited stock

Table 6: Length-composition negative log-likelihood by fishery in the O1 profile, at MFCL's scale and at log(B0) 3 higher (start-of-year growth).

fishery	name	n obs	at MFCL	at +3	change
3	3 LL DWFN 1c	54	1,370.6	1,354.3	−16.3
10	10 LL AU/NZ 1cd	27	412.3	407.1	−5.2
4	4 LL DWFN 1d	49	509.4	506.5	−2.9
8	8 LL DWFN 1cd	42	310.5	307.8	−2.7
7	7 LL AU/NZ 1ab	13	213.2	211.0	−2.2
6	6 LL PICT 1ab	31	671.3	670.2	−1.1
14	14 LL index 1abcd	57	162.9	162.0	−0.9
9	9 LL PICT 1cd	14	69.5	68.9	−0.6
12	12 TR 1d	7	268.3	268.2	−0.1
13	13 DN 1cd	2	10.6	10.6	−0.0
15	15 TR index 1ef	25	521.2	521.3	0.1
11	11 TR 1c	25	2,099.1	2,103.9	4.7

The longline fisheries drive the pull; the troll fishery 11 prefers MFCL's scale. With selectivity fixed, a lightly fished stock keeps more large, old fish, so opal's predicted longline catches are too small for the data.

MFCL predicts each length sample at its date. Its `length.fit` gives, for each sample, the predicted composition by age, whose mean lengths show length-at-age growing through the model year (Figure 4); opal predicts every sample from start-of-year length-at-age.

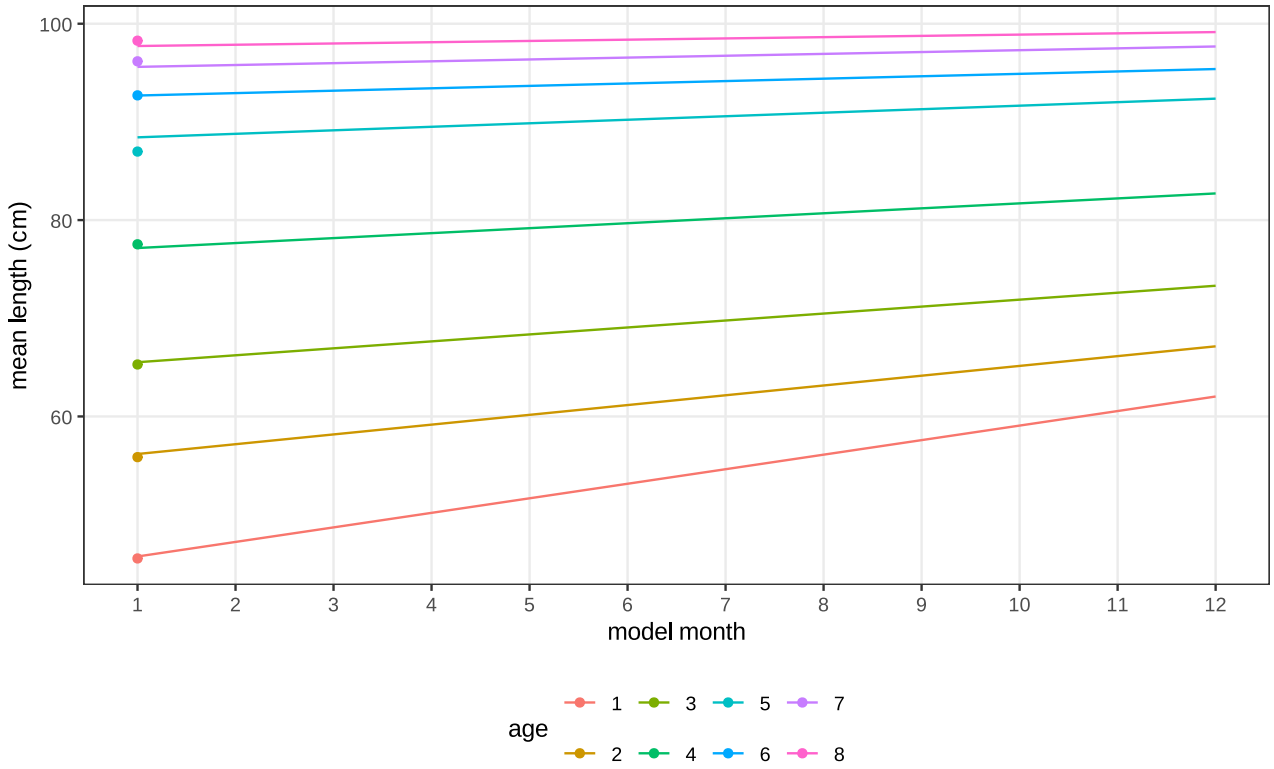


Figure 4: MFCL's mean length-at-age by model month (month 1 is October), from its predicted length compositions by age (M3), with opal's start-of-year length-at-age (points at month 1).

The test changes only opal's inputs: its growth fitted to MFCL's mid-year lengths (the average of model months 6 and 7). Figure 5 compares profiles over B0 for the annual model with start-of-year and mid-year growth, the same with fisheries 1-7 split into one fishery per season (Section 8), and the quarterly-age model.

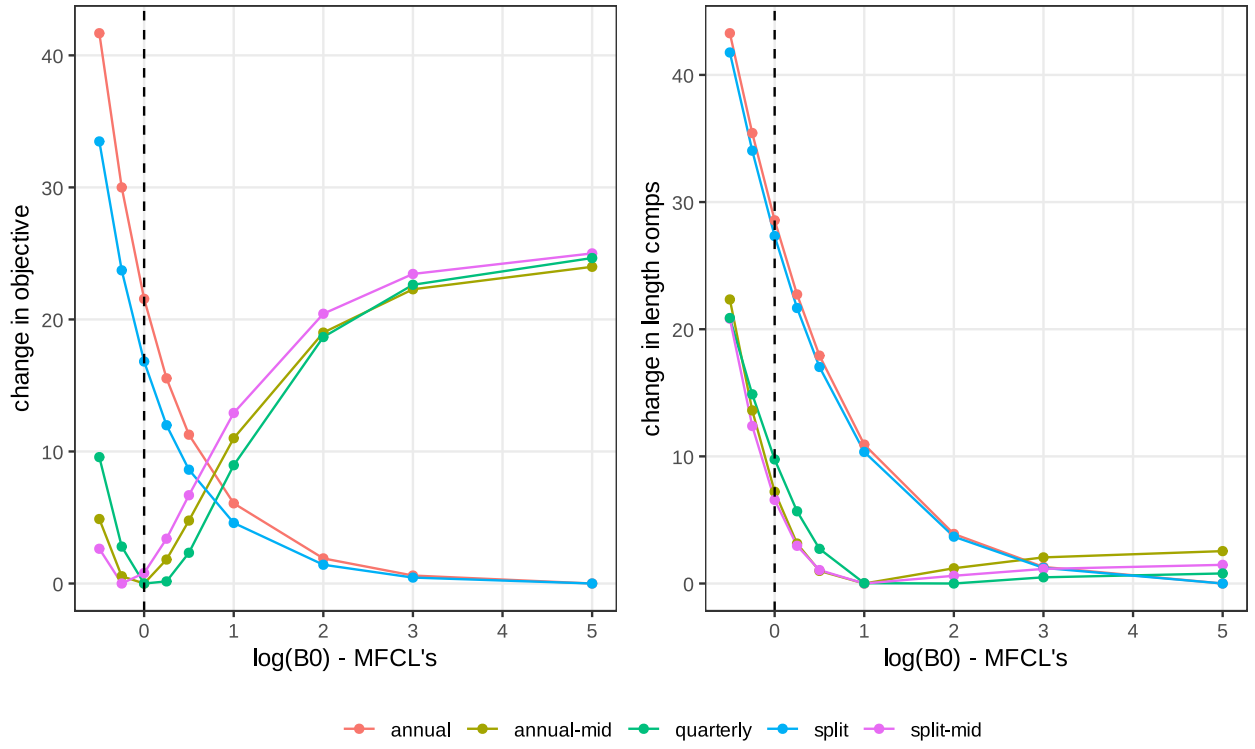


Figure 5: Profiles over log(B0) (relative to MFCL's): change in the total objective (left) and in the length-composition component (right). annual: O1′s set-up; -mid: growth at MFCL's mid-year lengths; split: one fishery per season for fisheries 1-7; quarterly: quarterly ages (B0 on the annual model's scale).

Table 7: Profile summaries: where the objective is smallest (log(B0) relative to MFCL's), how much it rises at +5, and depletion at MFCL's scale.

variant	best log(B0) - MFCL's	rise at +5	depletion at MFCL's scale
annual	5.00	0.0	0.755
annual-mid	0.00	24.0	0.614
split	5.00	0.0	0.752
split-mid	−0.25	25.0	0.608
quarterly	0.00	24.7	0.467

With mid-year growth the profile has a minimum near MFCL's scale and the index fit improves, and O1-mid (O1 with mid-year growth, B0 estimated) converges at log(B0) 13.95 against MFCL's 14.04. The length compositions on their own still favour a somewhat larger stock than MFCL's.

The quarterly-age model, which gives each quarter MFCL's length-at-age and selectivity and fits the length compositions by quarter, also has its minimum at MFCL's scale and fits the indices far better than any annual variant (CPUE negative log-likelihood 0.66 at MFCL's scale, against 15.70 for annual-mid and 44.66 for the O1 set-up). Its depletion at MFCL's scale (0.467) is lower than the annual variants', partly because its stock-recruitment curve is evaluated differently (Section 8); its B0 is not comparable with the annual models' beyond being a scale.

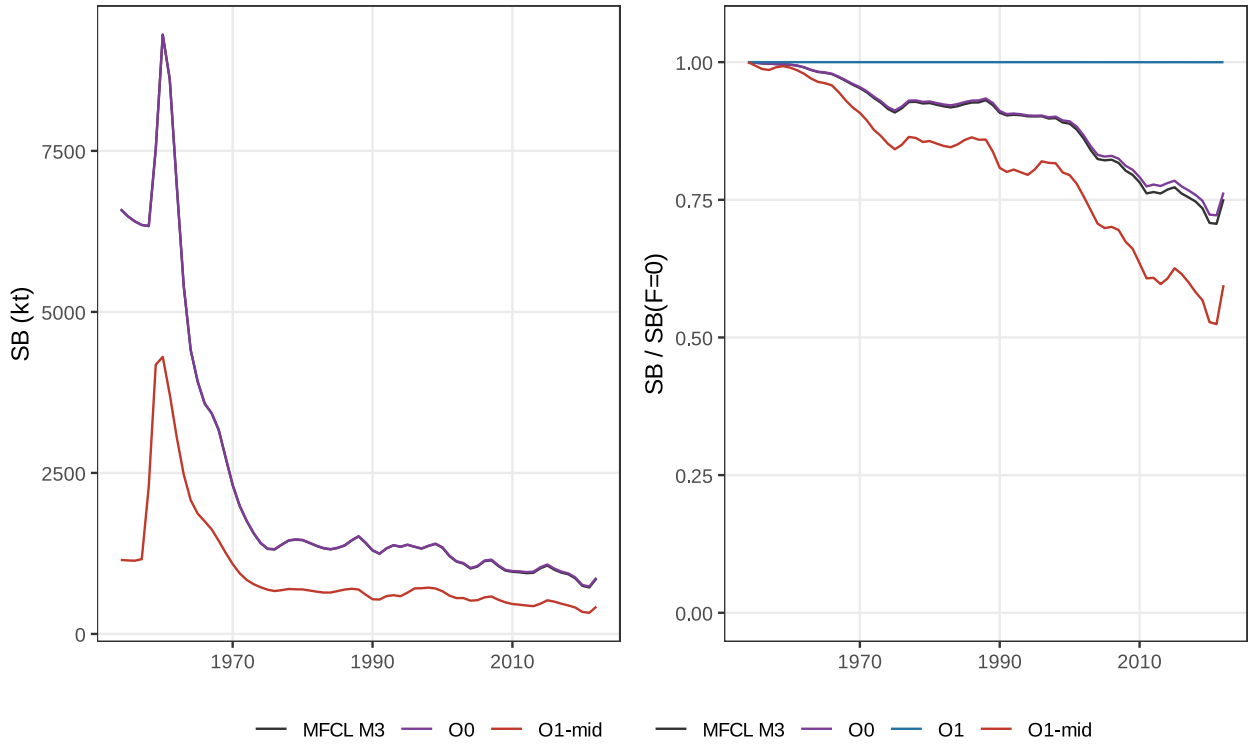


Figure 6: Spawning biomass and dynamic depletion: MFCL M3, O0, O1 and O1-mid. O1′s spawning biomass is off the scale.

8 Seasonal set-ups

Both set-ups use opal's inputs only.

Seasonal fisheries (`mfcl_split_seasonal_fisheries()`). Fisheries 1-7, whose MFCL selectivity differs by season, become four fisheries each (36 fisheries), each with one season's catch, selectivity and length compositions. This represents MFCL's seasonal selectivity exactly in the annual-age model and lets opal estimate seasonal curves; on its own it does not change the length-composition pull (Figure 5).

Quarterly ages (`mfcl_quarterly_ages()`). opal's time step and age classes become quarters: 48 quarterly ages and 276 steps, one season each. Recruitment stays annual: the recruitment deviations of the three quarters without recruitment are fixed at −30, and the initial numbers of quarterly ages that do not exist (cohorts recruit once a year) are near zero. Then:

- quarterly age $q = 4(a-1)+s$ holds annual age class a in season s only, so a selectivity by quarterly age is MFCL's seasonal selectivity (exactly, except for the plus group, which is present in every season);
- length-at-age, and weight at the time of catch, are MFCL's at each quarter's fishing month (opal's growth fitted to them, RMSE 0.68 cm);
- the length compositions are fitted by quarter, and the indices in the quarter they sample;
- M is MFCL's per quarter.

Table 8: Quarterly-age model at MFCL's values against MFCL M3.

quantity	rel. diff.
numbers at age, first year	1.7e-06
recruitment	5.9e-05
spawning biomass (max)	2.4e-02
spawning biomass (2022)	1.8e-02

The cost is the stock-recruitment relationship: opal treats every step as a recruitment, so with one recruitment in four steps its B0 is about four times the unfished spawning biomass and the curve is evaluated at a quarter of the usual relative stock size, which changes what steepness means. Depletion has to be taken against opal's unfished projection, and B0 treated as a scale parameter.

9 What opal would need

Table 9: Features of the MFCL model that opal does not have, how they were handled here, and their effect.

MFCL	opal	here
Two regions with movement; recruitment distributed over regions	one region	MFCL bridge to region 1 (M1, M2); scale doubles at each step
Conditional age-at-length likelihood	none	dropped with growth fixed (M3); scale doubles
Length compositions predicted at each sample's date, with within-year growth	start-of-year length-at-age; annual compositions	the main cause of O1′s unbounded scale; mid-year growth or quarterly ages fix it
Seasonal selectivity (fisheries 1-7)	selectivity by year	catch-weighted average; <0.1% on SB; seasonal fisheries or quarterly ages represent it
Catch-conditioned Baranov at each fishing incident's date (several per quarter for troll)	Pope removal at the start of each season	up to ~2% on SB (O0 against the replays)
Length-based cubic-spline selectivity, grouped across fisheries	logistic, double normal, double Richards; no mirrors (sharing via map)	MFCL's selectivity-at-age fixed (O0, O1)
VB growth with independent mean lengths at ages 2-4; SD log-linear in length	VB, Richards, cessation, age-specific K; SD or CV linear in length	age-specific K at ages 1-4 (0.5 cm RMSE)
Length compositions per incident, robust normal likelihood	annual compositions; multinomial, Dirichlet, Dirichlet-multinomial	summed by fishery and year; Francis divisors kept
Index abundance at the incident's date	start of the season	as opal (season-start numbers)
Recruitment from the previous year's spawning potential; last two at the mean	same year's spawning biomass; free deviations	MFCL's recruitment imposed (O0); last two deviations fixed (O1)

In order of their effect on this stock: length-at-age at the time of sampling (or a sub-annual time step with annual recruitment, which would also give seasonal selectivity, catch weight and index timing), then catch timing within the season, then multiple regions, then a conditional age-at-length likelihood for estimating growth.

10 Reproducing this document

From any working directory (`CLAUDE.md` has the R environment):

```
Rscript --no-init-file case-studies/swpo-alb/scripts/01-build-mfcl-bridge.r
Rscript --no-init-file case-studies/swpo-alb/scripts/02-compare-mfcl-bridge.r
```

```
Rscript --no-init-file case-studies/swpo-alb/scripts/03-convert-fit-swpo-alb-opal.r
Rscript --no-init-file case-studies/swpo-alb/scripts/04-length-comp-diagnostics.r
quarto render case-studies/swpo-alb/vignettes/swpo-alb-mfcl-to-opal.qmd
```

The MFCL fits, opal fits and profiles are cached (`SWPO_ALB_REFIT=true` repeats them). The document reads only `vignettes/data/`.

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